

"Gradient Boosting"

For Regression problem: First models prediction/output will be the mean of target column.

<u>IQ</u>	<u>cgpa</u>	<u>Salary</u>	<u>Predict 1</u>	<u>res1</u>
90	8	3	9.8	-1.8
100	7	4	9.8	-0.8
110	6	8	9.8	3.2
120	9	6	9.8	1.2
80	5	3	9.8	-1.8

Loss function of GB also known as pseudo residual.
= actual value - predicted value

we use Decision Tree as sub-model.

Step-1 Prediction will be the mean value of target.

Step-2 calculate residual

Step-3 For the next model input will be feature input and residual column (as target)

Step-4 Repeat all 3 steps and always target for model n will be the output of model $n-1$

For prediction: $\text{Pred} = m_1 + b_0 * m_2 + m_3 * b_0 + \dots$

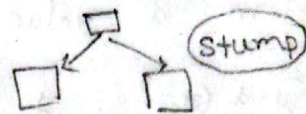
Step:5 now do the prediction and calculate the residual then send it to the next step.

Gradient-Boost

- ① weak learner $\rightarrow$ Decision Tree with max leaf 8-32
- ② learning rate / model weight $\rightarrow$ same for all model

Adaboost

Decision Tree with 2 leaf node



Different model weights.

Mathematical Intuition

Input training data set $\{(x_i, y_i)\}_{i=1}^n$, a differentiable loss function $L(y, F(x))$, number of iteration.

Step-1 initialize $f_0(x) = \arg \min_{\gamma} \sum_{i=1}^N L(y_i, \gamma)$
(gamma) $\rightarrow$

Step-2 For $m = 1$ to M : $\rightarrow$ num of decision tree

① For $i = 1, 2, \dots, N$ compute

residual $\rightarrow r_{im} = - \left[\frac{\partial L(y_i, f(x_i))}{\partial f(x_i)} \right]_{f=f_{m-1}}$

② Fit a regression tree to the targets r_{im} giving terminal regions R_{jm} , $j = 1, 2, 3, \dots, J_m$

③ For $j = 1, 2, 3, \dots, J_m$ compute

$$\gamma_{jm} = \arg \min_{\gamma} \sum_{x_i \in R_{jm}} L(y_i, f_{m-1}(x_i) + \gamma)$$

④ update $f_m(x) = f_{m-1}(x) + \sum_{j=1}^{J_m} \gamma_{jm} I(x \in R_{jm})$

Step-3

output: $\hat{f}(x) = f_m(x)$

why we take -ve gradient?

→ cause we want to move F_{m-1} in the direction that decreases the empirical loss. and -ve gradient points to that direction where loss is minimum.

Step-1

$$J_0(x) = \arg \min_y \frac{1}{2} \sum_{i=1}^n (y_i - y)^2$$

using Least Square as loss function

find a value of gamma so that $J_0(x)$ gives lowest loss.

$$\begin{aligned} \frac{d J_0(x)}{dy} &= \frac{d}{dy} \left(\frac{1}{2} \sum_{i=1}^n (y_i - y)^2 \right) \\ &= \sum_{i=1}^n -(y_i - y) \end{aligned}$$

If we use other loss function this will be change

now

$$\begin{aligned} - \sum_{i=1}^n (y_i - y) &= 0 \\ \Rightarrow - \sum_{i=1}^n y_i + y \sum_{i=1}^n 1 &= 0 \end{aligned}$$

$$\Rightarrow n y = \sum_{i=1}^n y_i$$

mean of target column
so for model 1: $y = \text{mean of target}$

$$\Rightarrow y = \frac{1}{n} \sum_{i=1}^n y_i$$

$$\hat{y}_i = f(x_i)$$

residual for i-th row
using m-th model (D^T)

Step: 2a

$$r_{im} = - \left[\frac{\partial L(y_i, \hat{y}_i)}{\partial \hat{y}_i} \right]_{\hat{y} = \hat{y}_{m-1}}$$

-ve gradient

$$= - \left[\frac{\partial}{\partial \hat{y}_i} L(y_i, \hat{y}_i) \right]_{\hat{y} = \hat{y}_{m-1}}$$

$$= - \left[\frac{\partial}{\partial \hat{y}_i} \frac{1}{2} (y_i - \hat{y}_i)^2 \right]_{\hat{y} = \hat{y}_{m-1}}$$

no Σ cause we are finding r_{im} for 1 row not all

$$= - [(-1)(y_i - \hat{y}_i)]$$

$$r_{im} = y_i - \hat{y}_i \rightarrow \text{For each row residual is actual - pred}$$

Step: 2b

① Fit a DT

② now DT max leaf node count will make max leaf number of cuts in data known as terminal region

Step-2c calculate output for each terminal region (leaf).

find the minimum value of y for j -th leaf of model - m

$$y_{jm} = \arg \min_y$$

$$\sum_{x_i \in R_{jm}} L(x_i, f_{m-1}(x_i) + y)$$

do differentiation to find min value of y

j -th terminal region for model m

$$\frac{d}{dy} \left[\frac{1}{2} \sum_{i=1}^n \{y_i - (f(x_i) + y)\}^2 \right]$$

$$= \sum_{i=1}^n (y_i - f(x_i) - y) (-1) = \sum_{i=1}^n (-y_i - f(x_i) + y)$$

$$\therefore -\sum_{i=1}^n y_i - \sum_{i=1}^n f(x_i) + y \sum_{i=1}^n 1 = 0$$

$$\Rightarrow y = \frac{1}{n} \left[\sum \{y_i + f(x_i)\} \right]$$

Gradient Boosting for classification

For classification we use log loss function as loss function.

Stage-1 Take a simple model $\rightarrow$ For classification take log-odds.

$$\text{log-odds} = \log\left(\frac{\text{class-1}}{\text{class-0}}\right) \rightarrow f_0(x)$$

To calculate probability using log-odd

$$P = \frac{1}{1 + e^{-\text{log-odd}}}$$

after that find the residual ($\text{res} = \text{actual class} - P$)
Then train a DT Regressor with input cols and use res as target

For the prediction

as result of f_0 is log-odd and from f_1 to f_m leaf value is probability so can't add them
convert the leaf values of all model into log-odd.

$$\text{For each leaf node } \text{log odd} = \frac{\sum \text{Residual}}{\sum [\text{Prob} \times (1 - \text{Prob})]}$$

If we are calculating log odd for leaf a then consider the residuals and probability of these rows which falls into this leaf.

Now do prediction for all rows. for a row sum of all probability where this row belongs. example: for DT1 leaf node val + DT2 leaf val + and so on. By doing this we will get the log-odd for this row then convert that sum value of this row to probability.
Then calculate residual again and pass it to next model.

Problem setup:

• Given a dataset for training (x_i, y_i) for $i = 1, 2, \dots, n$

• Now we have to find an additive model

$$F_M(x) = F_0(x) + \sum_{m=1}^M \eta \gamma_m h_m(x)$$

where, $h_m(x) \rightarrow$ is the m -th weak learner (DT)

$\eta \in (0, 1]$ is the learning rate (shrinkage)

γ_m is a scalar multiplier chosen for step m .

$F_0(x)$ is the initial predictor (for MSE mean of target)

Step-2b Fit a weak learner $h_m(x)$ to predict the residuals

$$h_m(x) \approx r_m \left[\text{fit using } \{x_i, r_{im}\}_{i=1}^n \right]$$

$h_m(x)$ is a decision tree, it outputs a piecewise-constant function.

$$h_m(x) = \sum_{j=1}^J V_{jm} \mathbf{1}\{x \in R_{jm}\} \text{ where } R_{jm} \text{ are terminal}$$

region (leaves) and V_{jm} is the leaf value.

$J \rightarrow$ num of terminal node (leaves)

$R_{jm} \rightarrow$ the region (subset of feature space) corresponding to the j -th leaf of m -th tree.

$V_{jm} \rightarrow$ the predicted value of assigned to region R_{jm}

$\mathbf{1}\{x \in R_{jm}\}$ means

it equals 1 if the data point x falls into the region R_{jm} otherwise 0.

Now find the minimum value of γ_m

after fitting $h_m(x)$ we choose γ_m to minimize the empirical loss when adding $\gamma_m h_m(x)$

It rescaled the new DT $\Rightarrow h_m(x)$ so that we move along it just right amount to most reduce the loss.

$$\gamma_m = \arg \min_{\gamma} \sum_{i=1}^n L(y_i, F_{m-1}(x_i) + \gamma h_m(x_i))$$

$$J(\gamma) = \frac{1}{2} \sum_{i=1}^n \left\{ y_i - (F_{m-1}(x_i) + \gamma h_i) \right\}^2 \quad \boxed{\text{here } h_i = h_m(x_i)}$$

$$= \frac{1}{2} \sum_{i=1}^n \left[\{ y_i - F_{m-1}(x_i) \} - \gamma h_i \right]^2$$

$$= \frac{1}{2} \sum_{i=1}^n (r_i - \gamma h_i)^2$$

$$= \frac{1}{2} \sum_{i=1}^n (r_i^2 - 2\gamma h_i r_i + \gamma^2 h_i^2)$$

$$= \frac{1}{2} \left[\sum_{i=1}^n r_i^2 - 2\gamma \sum_{i=1}^n h_i r_i + \gamma^2 \sum_{i=1}^n h_i^2 \right]$$

$$\frac{\partial J(\gamma)}{\partial \gamma} \Rightarrow -\sum_{i=1}^n h_i r_i + \gamma \sum_{i=1}^n h_i^2 = 0$$

$$\Rightarrow \gamma = \frac{\sum_{i=1}^n h_i r_i}{\sum_{i=1}^n h_i^2} \Rightarrow \boxed{\gamma = \frac{r^T h}{h^T h}}$$

now calculate the value of each leaf. many implementation take the leaf value rather than calculating global minima for each leaf.

min val
of leaf
j for Region

$$V_j^* = \arg \min_v \sum_{i \in R_j} \frac{1}{2} \{y_i - (F_{m-1}(x) + v)\}^2$$

$$L(v) = \frac{1}{2} \sum_{i \in R_j} (r_i - v)^2 \rightarrow y \text{ independent}$$

so instead of fit the tree m with y we can fit on residual.

$$\frac{dL}{dv} = \sum_{i \in R_j} (r_i - v)(-1) = 0$$

$$\Rightarrow - \sum_{i \in R_j} r_i + v \sum_{i \in R_j} 1 = 0$$

$$\Rightarrow \boxed{V_j = \frac{1}{|R_j|} \sum_{i \in R_j} r_i} \rightarrow \text{mean residual of all points in leaf } R_j$$

Note: so in least square method the best leaf value is the mean of the target (residual) values.

now after completing m -th iteration update the model

$$F_m(x) = F_{m-1}(x) + \eta h(x) \text{ with leaf value } v^*$$